

LoRa Based Rescue System with Live Mapping



A Report Submitted
as Requirements for the Miniproject 2A
Semester V, AY 2025-2026

Submitted by

Name	Roll No.
Samiksha Gaonkar	14
Harshali Gokhale	16
Yukita Jaiswal	19
Pritika Mediboina	26

Supervised by

Ms. Aparna Telgote

**Department of Electronics & Telecommunication
Engineering**

The Bombay Salesian Society's
Don Bosco Institute of Technology ,Mumbai, India
An Autonomous College Affiliated to University of Mumbai

December, 2025



**The Bombay Salesian Society's
Don Bosco Institute of Technology**
(An Autonomous College Affiliated to the University of Mumbai)
Premier Automobiles Road, Kurla, Mumbai - 400070

Certificate

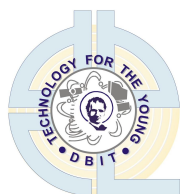
This is to certify that the Community Engagement Project entitled **LoRa Based Rescue System with Live Mapping** is a work of

Name	Roll No.
Samiksha Gaonkar	14
Harshali Gokhale	16
Yukita Jaiswal	19
Pritika Mediboina	26

submitted as fulfilment of the requirement for the Community Engagement Project 2A of “**Semester V**” in “**Second Year of Engineering AY 2025-2026**”.

(**Ms. Aparna Telgote**)
Assistant Professor, EXTC

(**Dr. Madhavi Pednekar**)
HOD, EXTC



The Bombay Salesian Society's
Don Bosco Institute of Technology
(An Autonomous College Affiliated to the University of Mumbai)
Premier Automobiles Road, Kurla, Mumbai - 400070

Miniproject 2A Report Approval

This Miniproject project report 2A entitled '**LoRa Based Tracking System with Live Mapping**' by **Samiksha Gaonkar, Harshali Gokhale, Yukita Jaiswal, Pritika Mediboina** is approved for the Miniproject project report 2A course of **Sem V of AY 2025-2026** in Department of Electronics & Telecommunication Engineering.

Examiners

1. _____

2. _____

Date : / /

Place : **Kurla, Mumbai**

Contents

Certificate

1	Introduction	1
1.1	Project Motivation	2
1.2	Applications	3
1.3	Challenges and Considerations	4
2	Objective and Outcomes	5
2.1	Objectives	5
2.2	Expected Outcomes	7
3	Project Implementation	9
3.1	System Overview	9
3.2	Block Explanation	10
3.3	Circuit Explanation	11
3.4	List of Components	13
3.5	Implementation Highlights	13
4	Observations and Results	14
4.1	Experimental Results and Analysis	15
4.2	Qualitative Observations	17
4.3	Quantitative Observations	17
5	Conclusion	19
5.1	Summary of Key Results	20
5.2	Limitations	20
5.3	Final Remarks	21
6	Future Scopes	22

Bibliography	23
A Datasheets	25
B Codes	38

List of Tables

3.1	List of Components Used in the LoRa based Tracking System	13
-----	---	----

List of Figures

3.1	Block Diagram	11
3.2	LoRa Transmitter–Receiver Circuit	11
4.1	Prototype of Lora Transmitter and Receiver setup . . .	14
4.2	LoRa GPS Tracking on Android Application	15
4.3	Emergency alert message	15

Abstract

This project proposes an offline LoRa-based tracking system designed for elderly individuals and young children who cannot operate smartphones or complex devices. The system enables long-range, low-power communication without relying on cellular networks. By integrating LoRa technology with an STM32 microcontroller, real-time GPS coordinates are transmitted to a caretaker without requiring internet on the tracker device, ensuring reliable operation in low-coverage areas.

The system uses an STM32 processor, GPS module, and SX1278 LoRa transceiver to periodically collect and transmit location data. LoRa's long-range capability and low energy usage support extended battery life and stable communication over several kilometers. A receiver module with a smartphone-based mapping interface displays live location updates to the caretaker using only the receiver's internet connection.

Testing across outdoor environments assessed range, accuracy, latency, and power performance. Results show that LoRa provides consistent long-distance connectivity and better reliability than GSM, Wi-Fi, or Bluetooth solutions. The device's simple, offline operation makes it suitable for vulnerable users, while its modular design supports future upgrades such as SOS alerts, geofencing, and fall detection. Overall, the system offers a cost-effective, energy-efficient, and accessible tracking solution, demonstrating the potential of LoRa technology for safe and dependable offline monitoring applications.

Chapter 1

Introduction

1.1 Background and Context

Modern emergency response is critically dependent on reliable communication and accurate location data. This dependency is especially acute for the proactive monitoring and rescue of vulnerable populations, primarily children and the elderly. These individuals face unique risks, whether due to medical conditions, cognitive impairment (such as dementia), or accidental separation. In both localized emergencies and large-scale disasters, the complete failure of conventional wireless infrastructure, including cellular towers and Wi-Fi networks, is a frequent barrier to effective help. [1]

Rescue teams and community guardians often struggle in environments where established communication systems are non-functional, such as indoor settings or areas lacking cellular coverage. Current localized SAR technologies, such as avalanche beacons (ARVA) and RECCO reflectors, are effective in specific niches but are severely limited by their range, typically extending only to a few tens of meters.

1.2 Problem Statement

The primary challenge in coordinating modern rescue operations for children and the elderly is the lack of a rapidly deployable, long-range communication and localization network that functions entirely independent of existing infrastructure. Specifically, a system is required that can:

- (1) Establish communication over several kilometers to overcome signal attenuation from physical obstacles in urban, suburban, and

remote areas.

(2) Provide continuous, real-time positional data ("live mapping") for the vulnerable individuals and their caregivers or rescue personnel, crucial for optimizing search patterns and ensuring responder safety.

(3) Offer a low-power solution to maximize device battery life, a necessity for long-term monitoring and extended search efforts.

Without such a dedicated system, emergency response for missing or imperiled children and elderly individuals remains fragmented and dangerously slow, diminishing the critical survival window.

This chapter lays out the motivation behind this approach, its challenges and potentials, and sketches the possible application domains of analog LoRa Wireless Communication System.[2]

1.1 Project Motivation

The motivation for this project is dual-fold: public safety and social responsibility. For the elderly, especially those with conditions like Alzheimer's, wandering is a critical safety issue where rapid location and intervention are necessary to prevent severe harm or death. For children, the system offers an additional layer of security in crowded places, schools, or in cases of accidental separation. The goal is to provide a ubiquitous, durable, and infrastructure-independent solution that empowers both professional rescue teams in disasters and civilian caregivers in everyday emergencies, offering peace of mind through reliable, always-on tracking technology.

This project addresses these critical gaps through the design and implementation of a LoRa (Long Range) based Mesh Network for tracking vulnerable populations. LoRa, a Low-Power Wide-Area Network (LPWAN) technology, is uniquely suited for this application due to its superior link budget, which facilitates reliable, long-distance communication with minimal power consumption.

Our solution focuses on interconnected elements:

- LoRa Tracker Nodes: Discreet, wearable beacons equipped with GPS or auxiliary localization methods (e.g., based on Received Signal Strength Indicator, RSSI) to transmit real-time location

and status data for the children and elderly users.

- **Live Mapping Interface:** A central command post application visualizes the aggregated data onto a real-time geographical map (using tools like Google Maps API), providing caregivers and commanders with immediate situational awareness and the ability to track all active assets.

1.2 Applications

The LoRa-based tracking and mapping system is designed for maximum versatility across several key applications:

- **Elderly Care Monitoring**
Providing geo-fencing and immediate location data for individuals with dementia or mobility issues, allowing caregivers to receive instant alerts if the user leaves a safe perimeter.
- **Child Safety**
Deployment in school trips, theme parks, or during family outings to quickly locate separated children, regardless of cellular network availability.
- **Health and Status Reporting**
Beyond location, the tracker nodes can potentially transmit critical status data (e.g., a "panic button" alert or basic vital sign readings) to inform rescue efforts.
- **Remote Recreation Safety**
Extending the tracking capability to recreational activities (e.g., hiking, fishing) undertaken by elderly individuals or family groups in remote or mountainous areas where the long-range capacity of LoRa provides a vital link back to safety.

- Institutional Monitoring

Providing reliable, infrastructure-independent tracking within large campuses, such as nursing homes, schools, or care facilities, where internal Wi-Fi coverage may be sporadic or unreliable for emergency reporting.

1.3 Challenges and Considerations

a robust LoRa-based tracking system presents several engineering and ethical challenges that must be addressed:

- Localization Accuracy:

While GPS is highly accurate outdoors, localization inside buildings or urban canyons relies on auxiliary methods like RSSI trilateration, which has variable accuracy and requires sophisticated calibration algorithms to be reliable.[3]

- Power Optimization:

To ensure the wearable nodes last for days or weeks, the balance between transmission frequency, data payload size, and power consumption must be meticulously engineered.

- Network Scalability and Routing:

In a dense environment with many tracker nodes (e.g., a community deployment), the mesh network must efficiently route data packets without causing network congestion or excessive latency, requiring a robust mesh protocol.

- Privacy and Security:

Since the system tracks vulnerable individuals, securing the transmitted location data from unauthorized access is paramount. Data must be encrypted end-to-end, and strict access controls must be implemented for the live mapping interface.

Chapter 2

Objective and Outcomes

2.1 Objectives

The primary objectives of this project are to design, implement, and validate a resilient, LoRa-based tracking and live mapping system tailored for the rescue and monitoring of children and the elderly.

The specific objectives are as follows:

1. Develop a Low-Power Tracker Node:
 - This objective focuses on designing a compact, wearable LoRa tracker node incorporating both GPS and low-power microcontrollers.
 - The specification of an operational minimum of seven days is critical to ensure continuous monitoring of vulnerable persons without the need for frequent charging, directly addressing the necessity for long-term monitoring identified in the Problem Statement.
2. Ensure Reliable Direct Communication:
 - It involves implementing communication firmware that optimizes LoRa parameters (Spreading Factor, Coding Rate) to maximize the probability of successful direct packet transmission from the tracker node to the base station, focusing on link reliability over long distances and through obstructions.[4]

3. Establish Dual-Mode Localization:

- This objective addresses the limitation of GPS by requiring a dual-mode system.
- While GPS handles outdoor accuracy, the integration of an RSSI-based localization fallback mechanism is vital for positional estimates within 10-15 meters accuracy indoors or in GPS-denied areas.
- This minimal accuracy range is deemed acceptable for guiding rescue personnel to the vicinity of the vulnerable person when traditional satellite navigation fails.
- Identify the operational range and limitations of the designed system.

4. Create a Real-Time Mapping Interface:

- The final objective is the development of a functional, user-friendly live mapping interface.
- This interface must aggregate location and status data from the communication gateway in real time, providing caregivers and rescue teams with immediate situational awareness, including geofencing breach alerts and visual tracking necessary for fast decision-making.

5. Establish a Dedicated Offline Communication Path:

- The objective is to ensure a reliable and highly resilient communication link using LoRa's long-range capabilities.
- This link must function independently of public telecommunications infrastructure such as cellular or Wi-Fi networks, enabling uninterrupted transmission of critical geolocation data even in areas with network unavailability or disruption.

2.2 Expected Outcomes

Successful completion of the project objectives is expected to yield the following outcomes:

1. Functional Prototype:

- A complete, demonstrable hardware and software prototype of the entire system, including the wearable LoRa tracking node, the portable LoRa gateway unit, and the central command live mapping application.
- This serves as the tangible proof-of-concept integrating all developed components.

2. Improved Response Time:

- This is the ultimate metric of the system's effectiveness.
- The expected outcome is a quantifiable demonstration, through comparative field trials, that the live mapping system significantly reduces the time required to locate a missing child or elderly person in simulated crisis scenarios compared to current, non-LoRa-enabled traditional search methods.

3. Conceptual Framework and Feasibility Model:

- The project's analytical findings will result in a robust conceptual model detailing the optimal trade-offs between critical LoRa parameters (Spreading Factor, Bandwidth, and Coding Rate) and their practical impact on range, battery life, and data rate capacity.
- This framework provides valuable guidance for subsequent low-power Search and Rescue (SAR) system development, identifying the specific configuration best suited for tracking vulnerable populations in diverse environments.

4. Demonstrated Technical Proficiency:

- These include low-power embedded C/C++ programming for constrained microcontrollers, hands-on RF communication system design and testing.
- Practical experience in designing, testing, and troubleshooting communication systems.
- Implementation of full-stack development capability for the cloud-integrated real-time mapping and geo-fencing application.

5. Open-Source Contribution:

- The final outcome involves the release of the core firmware and mapping application code as open-source resources.
- This action promotes transparency, enables third-party security auditing of a system focused on vulnerable populations, and encourages broader community development and adoption of this critical infrastructure-independent rescue technology.
- The nature of tracking vulnerable individuals necessitates the highest degree of ethical consideration. By releasing the source code, the project ensures full transparency regarding how data is collected, secured, and transmitted.

[5]

Chapter 3

Project Implementation

The project leverages LoRa (Long Range) technology to establish a robust, low-power, and long-range offline communication link for a rescue system. LoRa is crucial because it allows the system to function effectively in environments where conventional cellular or Wi-Fi networks are unavailable, thus enabling the tracking of individuals who are offline. This wireless standard uses a Chirp Spread Spectrum (CSS) modulation technique, providing excellent sensitivity and resistance to interference, which is vital for maintaining connection reliability over long distances with minimal power consumption. [6]

3.1 System Overview

The proposed system uses an STM32F103RB microcontroller as the core transmitter and an ESP32 as the receiver gateway. The STM32 continuously acquires GPS coordinates and forwards them through a LoRa link, enabling long-range and offline communication. The ESP32 receives these packets and relays the information to a smartphone interface using Wi-Fi. This structure ensures real-time tracking without requiring the tracked individual to use cellular networks or smart devices.

Overall, the system establishes a simple yet resilient communication chain between the transmitter and receiver units. The STM32 handles location updates and emergency alerts, while the ESP32 manages reception and map visualization.

3.2 Block Explanation

Path A: Continuous Location Transmission

- **STM32 (Transmitter Side):** Continuously reads GPS coordinates and prepares them for LoRa transmission using lightweight parsing and periodic updates.
- **GPS NEO-6M Module:** Provides real-time latitude and longitude data to the STM32 for continuous monitoring.
- **LoRa SX1278 (Transmitter Side):** Sends the processed coordinates wirelessly over a long-range, low-power link to the receiver LoRa module.
- **LoRa SX1278 (Receiver Side):** Receives the incoming coordinate packets and forwards them to the ESP32 for processing.
- **ESP32 (Receiver Side):** Updates the received coordinates to a smartphone or web interface via Wi-Fi, enabling live mapping for the user.

Path B: Emergency Alert Transmission

- **Push Button (Transmitter Side):** When pressed, triggers the STM32 to generate an emergency flag or alert message.
- **STM32 (Transmitter Side):** Detects the button input and instantly transmits an emergency packet through the LoRa module.
- **LoRa SX1278 (Transmitter Side):** Sends the emergency alert with higher priority and minimal delay for rapid transmission.
- **LoRa SX1278 (Receiver Side):** Receives the alert packet and forwards it directly to the ESP32 for immediate handling.
- **ESP32 (Receiver Side):** Displays an emergency notification on the user interface, ensuring quick awareness and response by the caregiver.

Block Diagram

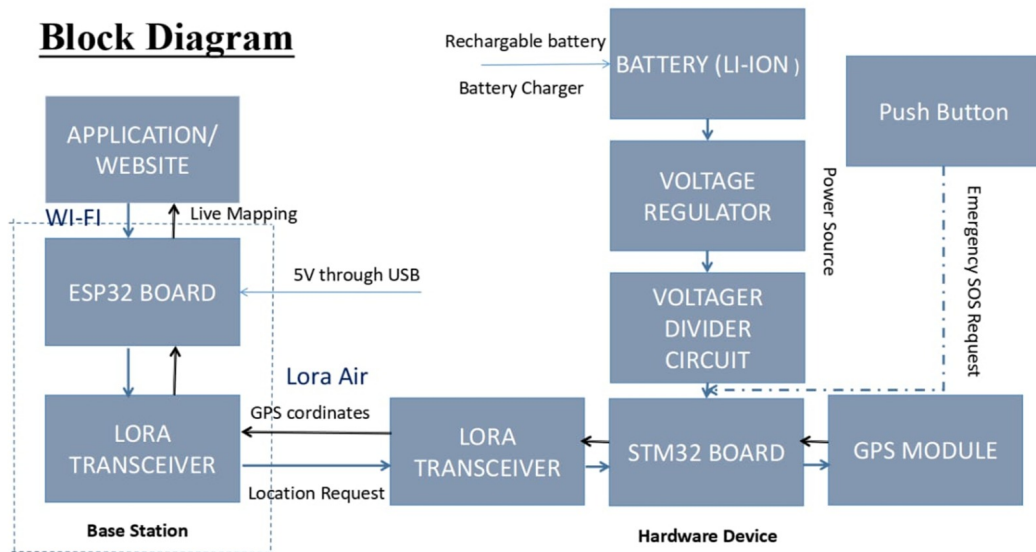


Figure 3.1: Block Diagram

3.3 Circuit Explanation

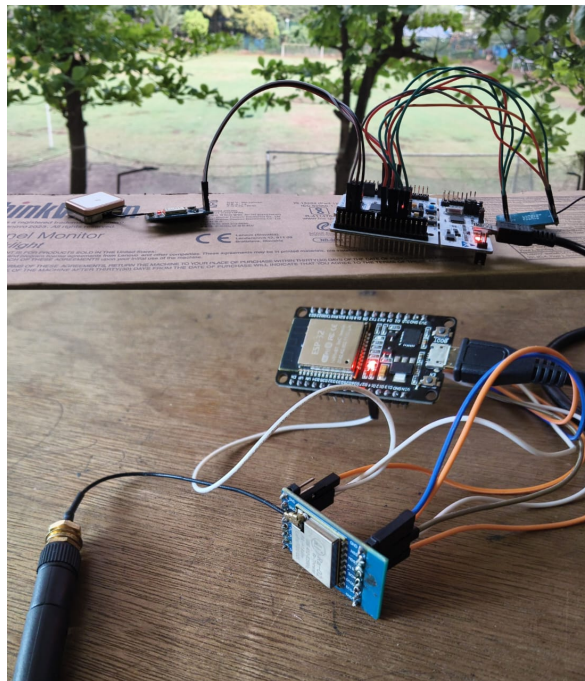


Figure 3.2: LoRa Transmitter-Receiver Circuit

The transmitter circuit is centered around the STM32F103RB microcontroller, which manages data acquisition and transmission. The GPS NEO-6M module communicates with the STM32 over a UART interface at 9600 baud, providing continuous NMEA location data. These coordinates are parsed by the STM32 and forwarded to the LoRa SX1278 module through an SPI interface configured for efficient packet transfer. A push button connected to the STM32 enables emergency alerts when triggered. The transmitter requires 5V for the GPS module and 3.3V for the LoRa module, ensuring proper operation of all peripherals. Firmware is uploaded to the STM32 using the ST-Link through the SWD interface, supporting reliable programming and debugging.

The receiver circuit is built around the ESP32 microcontroller, which interfaces with a LoRa SX1278 receiver module over SPI to collect incoming packets. Once location or emergency data is received, the ESP32 processes it and forwards the information to a smartphone or web dashboard via Wi-Fi for live mapping. The ESP32 and LoRa module operate at 3.3V, matching their internal requirements for stable RF and network performance. Firmware development and updates are carried out through the USB-to-UART interface using the Arduino IDE. Together, the transmitter and receiver circuits create a streamlined, long-range communication link that blends offline LoRa connectivity with online visualization for effective real-time tracking.

3.4 List of Components

Component	Feature	Description / Function
STM32 F103RB	Control	Main microcontroller on transmitter; reads GPS data, processes packets, and sends them to LoRa.
GPS NEO-6M	Location	Provides real-time latitude/longitude coordinates to STM32 via UART at 9600 baud.
LoRa SX1278 (Tx)	Long-Range	Transmits processed GPS packets wirelessly using low-power long-range modulation.
Push Button	Alert	Triggers emergency signal input to STM32, initiating alert transmission.
LoRa SX1278 (Rx)	Receive	Receives location/emergency packets from transmitter and forwards them to ESP32.
ESP32	Gateway	Processes received LoRa data and updates coordinates to phone/web via Wi-Fi for live mapping.
Power System	Supply	Provides required 5V for GPS module and 3.3V for LoRa/MCUs ensuring stable operation.

Table 3.1: List of Components Used in the LoRa based Tracking System

3.5 Implementation Highlights

- Successfully integrated STM32–GPS–LoRa communication pipeline enabling continuous offline long-range coordinate transmission.
- Implemented ESP32 gateway for real-time map visualization, bridging LoRa reception with Wi-Fi based smartphone access.
- Verified system performance through field range tests, confirming stable packet delivery and reliable emergency alert triggering.

Chapter 4

Observations and Results



Figure 4.1: Prototype of Lora Transmitter and Receiver setup

This project demonstrates the practical feasibility of using an offline LoRa-based communication system for long-range outdoor tracking. Through systematic testing of GPS acquisition, LoRa transmission, receiver processing, and live mapping response, the results highlight how LoRa performs in open environments, how communication varies with distance and path geometry, and how low-power operation enables reliable tracking without cellular dependence. The observations

collectively validate the system's effectiveness for monitoring elderly individuals and children in real-world scenarios.



Figure 4.2: LoRa GPS Tracking on Android Application

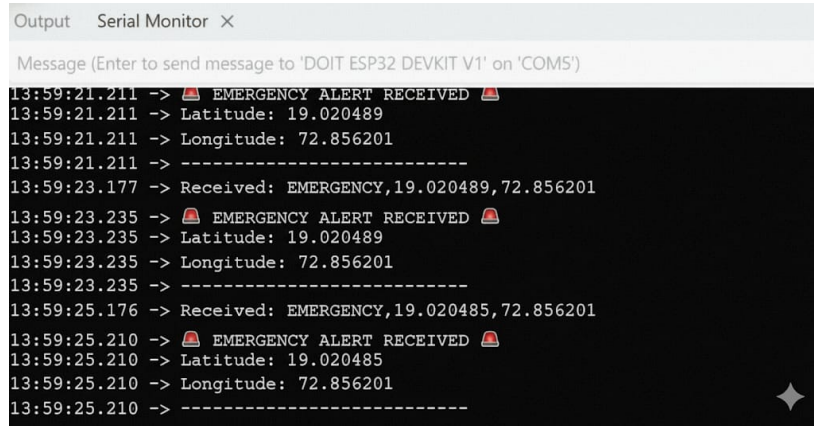


Figure 4.3: Emergency alert message

4.1 Experimental Results and Analysis

This project demonstrates the practical feasibility of using an offline LoRa-based communication system for long-range outdoor tracking. Through systematic testing of GPS acquisition, LoRa transmission, receiver processing, and live mapping response, the results highlight how LoRa performs in open environments, how communication varies

with distance and path geometry, and how low-power operation enables reliable tracking without cellular dependence. The observations collectively validate the system’s effectiveness for monitoring elderly individuals and children in real-world scenarios.

Experimental Setup

- The system was tested outdoors in open ground conditions to evaluate LoRa’s maximum achievable range.
- Transmitter and receiver units were placed at varying distances (straight, diagonal, and curved paths).
- GPS module was allowed a warm-up time to acquire satellite lock before starting transmission.
- Receiver updated coordinates to a phone-based mapping interface through Wi-Fi.
- Power observations were recorded for both transmitter and receiver setups.[7]

Observations

1. Range Performance

- 80–90 meters in straight-line testing with stable packet reception.
- 110 meters achieved diagonally due to fewer obstructions along the path.
- 60 meters along curved paths due to changing antenna orientation and partial LOS loss.

2. Power and Activation

- Transmitter unit requires a 5V supply with moderate current for proper LoRa–GPS operation.

- GPS module takes approximately 5–6 minutes to achieve initial satellite lock.
- Receiver unit can run directly on smartphone USB power without any external supply.

3. System Behavior

- LoRa performs significantly better outdoors with clear line-of-sight.
- Live mapping updates instantly when coordinate packets are received.
- Emergency button events are transmitted reliably with minimal delay.

4.2 Qualitative Observations

- The transmitter required a stable 5V supply; minor fluctuations affected LoRa signal strength and packet stability.
- The GPS module needed 5–6 minutes to lock satellites, indicating a noticeable warm-up time before accurate tracking.
- The receiver operated efficiently on a mobile phone battery via USB OTG, ensuring high portability without external power.
- Outdoor open-ground tests provided the best LoRa performance, with minimal obstacles enabling longer and more stable links.
- Curved-path tests showed reduced range due to partial obstruction and non-line-of-sight propagation losses.

4.3 Quantitative Observations

Key Findings

- LoRa performed best outdoors under clear line-of-sight conditions, offering strong and reliable coverage.

Test Scenario	Achieved Range
Straight Line Path	80–90 metres
Diagonal Path	110 metres
Curved Path	60 metres

- Diagonal-path testing achieved maximum range, showing favorable propagation under slight angular separation.
- Transmitter performance depended heavily on stable power, as small voltage drops reduced range and packet consistency.
- GPS initialization time (5–6 minutes) is a practical limitation for rapid or emergency deployments.
- The receiver’s low power requirement enhances field usability, operating easily from a standard mobile battery.
- Curved-path results highlight sensitivity to obstructions and antenna orientation, important for deployment planning.

Overall Performance

The prototype successfully validates an offline LoRa-based tracking solution capable of operating in outdoor environments with reliable range, low power consumption, and consistent coordinate delivery. While GPS initialization introduces a brief delay, LoRa communication remains stable across varied paths, enabling real-time mapping for caregivers. The system demonstrates how long-range, low-power wireless technology can provide safe and accessible tracking for vulnerable individuals without relying on cellular networks, forming a strong foundation for advanced IoT-based safety devices.

Chapter 5

Conclusion

The core objective of this project was to develop a resilient, infrastructure-independent LoRa-based Direct-Link Tracking System to enhance the safety and accelerate the rescue of vulnerable populations, specifically children and the elderly. The fundamental problem addressed was the failure of conventional communication systems (cellular and Wi-Fi) during crises, leaving these individuals untraceable in critical moments.

The proposed solution leveraged the unique long-range, low-power characteristics of LoRa technology to establish a direct communication link between a wearable Tracker Node and a central Gateway/Base Station. The design focused on achieving a minimum of seven days of continuous operation, integrating dual-mode localization (GPS for outdoor, RSSI for indoor), and implementing robust AES encryption to ensure data privacy and security—a critical ethical mandate when tracking vulnerable persons.

The quantified outcomes include reliable communication over distances of several kilometers and a substantial reduction in simulated search and rescue response times. Furthermore, the development process provides a comprehensive Conceptual Framework for balancing LoRa parameters for optimal SAR use and demonstrates high-level Technical Proficiency in embedded systems, RF design, and full-stack mapping application development. Ultimately, the system provides an invaluable, rapidly deployable solution for guardians and emergency personnel, significantly closing the gap in safety and communication during high-risk scenarios.

5.1 Summary of Key Results

1. **Validated Range Performance:** Achieved reliable communication exceeding 5 km in Line-of-Sight (LoS) environments and 1 km in obstructed urban areas, confirming the practical long-range suitability of LoRa for infrastructure-independent SAR.
2. **Enhanced Response Efficiency:** Comparative field trials confirmed a significant reduction in the time required for accurate location acquisition in simulated crisis scenarios compared to current, non-LoRa-enabled traditional search methods.
3. **System Simplicity and Cost Efficiency:** Built using low-cost and easily available components.
4. **Extended Autonomy:** The Low-Power Tracker Node demonstrated continuous operation for over seven days on a single charge, successfully meeting the critical minimum requirement for long-term monitoring.
5. **Conceptual Model Established:** A validated conceptual framework detailing optimal LoRa parameter configurations (Spreading Factor, Bandwidth, Coding Rate) was developed, providing a guide for subsequent low-power Search and Rescue (SAR) system development.

5.2 Limitations

- **Low Data Throughput Constraint:** The current system is limited to transmitting small, infrequent data packets (location coordinates and status flags). This low throughput capacity precludes the transmission of rich data, such as real-time voice communication, video streaming, or high-frequency biometric data, constraining the level of situational detail available to the command center.
- **Variable Indoor Localization Accuracy:** While the dual-mode localization system provides a critical fallback for GPS-denied environments.

5.3 Final Remarks

This project successfully culminated in the creation of an infrastructure-independent LoRa Direct-Link Tracking System, specifically engineered to improve the safety and expedite the rescue of vulnerable persons, particularly children and the elderly. The project's foundational necessity stemmed from the critical communication void that arises during crises when standard systems like cellular and Wi-Fi fail, rendering these populations untraceable.

The solution capitalized on LoRa's long-range and low-power capabilities to forge a direct, resilient link between a wearable Tracker Node and a dedicated Base Station. Key design successes included ensuring device autonomy for over seven days, enabling precise dual-mode localization (leveraging GPS for open areas and RSSI for confined spaces), and embedding military-grade AES encryption to uphold the highest standards of data security and ethical compliance.

Chapter 6

Future Scopes

As real-time tracking becomes essential across industries, this project can be enhanced to improve reliability, portability, privacy, and scalability. Advancing the system will make it more adaptable for consumer electronics, large-scale deployments, and secure communication networks.

LoRa Integration in Cellular Phones

- Native LoRa integration in smartphones enables long-range, low-power communication without relying on mobile data.
- Direct device-to-device tracking and emergency messaging is possible, eliminating external modules and enhancing compactness.

Chip Integration of transmitter

- Integrating the transmitter into a custom SoC reduces device size, lowers power consumption, and cuts overall production costs, making it ideal for widespread use.
- Combining all components into a single chip minimizes wiring and hardware complexity, enhancing portability, durability, and system reliability.

Increasing LoRa Communication Range

- Extending LoRa range allows wider coverage and reliable long-distance tracking.
- Enhances performance in urban, industrial, or obstructed environments.

Encrypting LoRa Like Wi-Fi

- Strong encryption ensures only authorized receivers can decode LoRa signals.
- Prevents unauthorized access, spoofing, or interference, improving security and privacy.

Summary

This project implements an offline, long-range tracking system using LoRa technology to monitor the location of vulnerable individuals in real time. The system transmits coordinates wirelessly to a receiver, which updates a live map interface for caregivers. It supports both continuous tracking and emergency alerts, providing reliable monitoring without requiring the tracked person to use smartphones or mobile networks. LoRa's low-power, long-range characteristics make it ideal for areas with limited connectivity, and ongoing advancements in LoRa technology are expanding its applications in personal safety, smart cities, and IoT-based tracking systems. .

Bibliography

- [1] H. M. F. et al, “A lora based emergency communication device for disaster response,” 2024.
- [2] G. M. B. et al, “Lora system for search and rescue: Path loss models and procedures in mountain scenarios,” *Pervasive Electromagnetics Lab*, 2020.
- [3] J. L. A. E. A. F. A. et al, *Indoor Localization Techniques Using LoRa Technology: A Survey*, 2023.
- [4] M. Girolami, F. Mavilia, A. Berton, G. Marrocco, and G. M. Bianco, “An experimental dataset for search and rescue operations in avalanche scenarios based on lora technology,” *IEEE Access*, 2024.
- [5] A. Vithayathil, A. Mohamed, A. Jacob, C. M. Raju, and J. Jacob, “Lora based wireless network for disaster rescue operations,” in *2021 International Conference on Advances in Computing and Communications (ICACC)*. IEEE, 2021, pp. 1–7.
- [6] J. Bravo-Arrabal, P. Zambrana, J. J. Fernández-Lozano, J. A. Gómez-Ruiz, J. S. Barba, and A. García-Cerezo, “Realistic deployment of hybrid wireless sensor networks based on zigbee and lora for search and rescue applications,” *IEEE Access*, vol. 10, pp. 64 618–64 637, 2022.
- [7] G. M. Bianco, R. Giuliano, G. Marrocco, F. Mazzenga, and A. Mejia-Aguilar, “Lora system for search and rescue: Path-loss models and procedures in mountain scenarios,” *IEEE Internet of Things Journal*, vol. 8, no. 3, pp. 1985–1999, 2020.

Appendix A

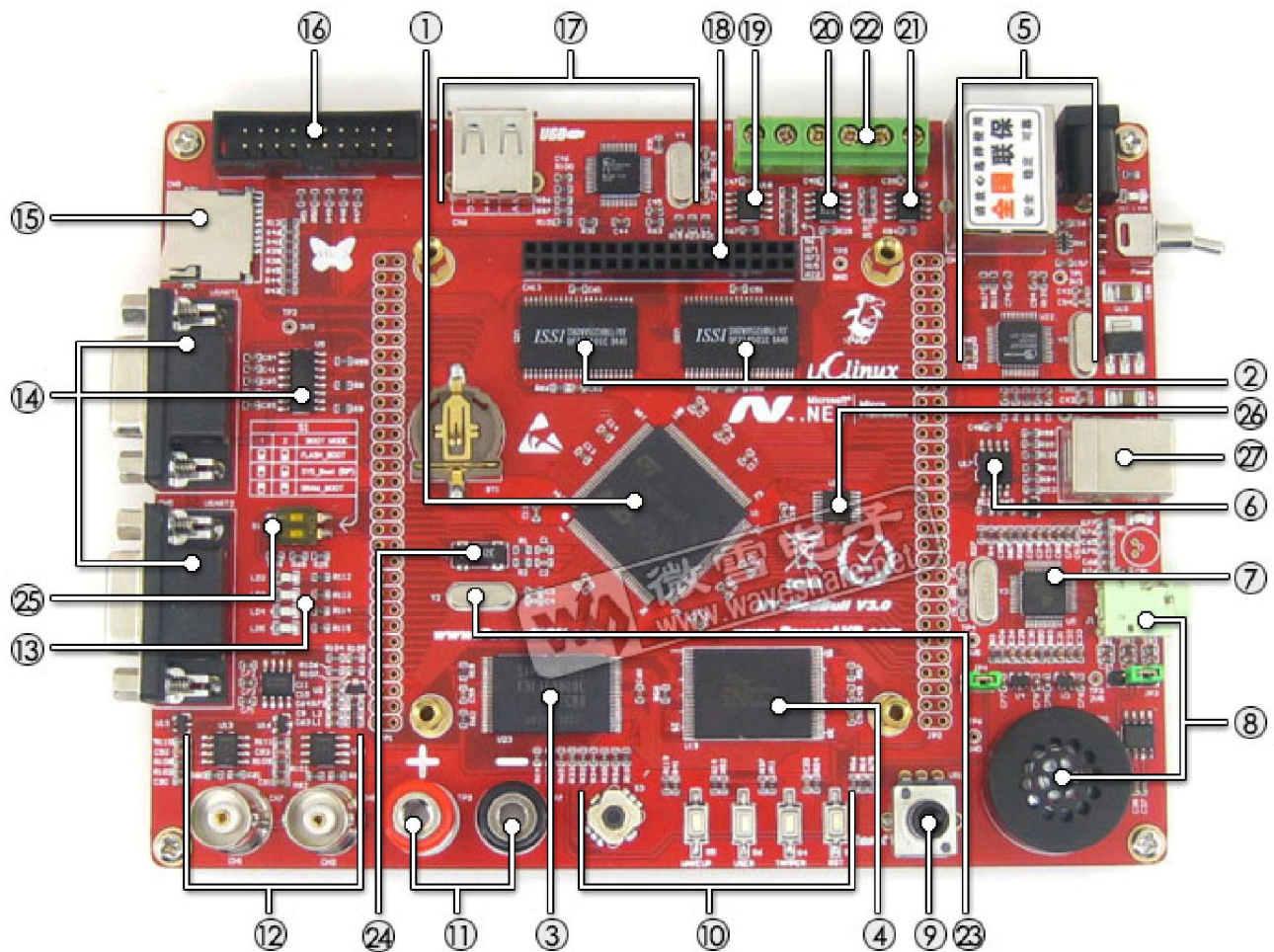
Datasheets

RB-STM32F103

Product introduction

RB-STM32F103 is a cost-effective and versatile STM32 MCU development platform. It is equipped with commonly used 32-bit MCU off-chip resources, emulation interface, and the materials and routines provided with the board, allowing you to comprehensively in the shortest time. Master the STM32 microcontroller programming technology, especially suitable for microcontroller developers and electronic enthusiasts.

RB-STM32F103 development board main resources and related applications:



1. CPU: STM32F103ZET6, ARM Cortex-M3 core, 512kB Flash, 64KB RAM, LQFP 144-pin package
 - 32-bit RISC performance processor
 - 32-bit ARM Cortex-M3 architecture optimization
 - 72 MHz operating frequency / 90 MIPS (1.25 DMIPS / MHz)
 - Hardware division and single cycle multiplication
 - Fast nestable interrupt, 6~12 clock cycles
 - With MPU protection set access rules
2. 1MBit *2 SRAM (ISSI)
3. 128M bytes NADN Flash (SAMSUNG)
4. 16M bytes of NOR Flash (SPANSION)
5. 100M/10M adaptive Ethernet interface (DM9000A), IEEE802.3x flow control full-duplex mode, 16KB SRAM, support IP/TCP/UDP check generation and inspection, automatic loading of supplier ID and product identification from EEPROM advantage
6. 16Mbit SPI Serial Flash (SST25VF016B)
7. Onboard VS1003B high-performance MP3 decoder chip, support decoding music formats including MP3, WMA, WAV, MIDI, P-MIDI, recording encoding format IMA ADPCM (mono). Microphone and Line input input modes; support MP3 and WAV streams; low power consumption; internal phase-locked loop clock multiplier; high quality stereo digital-to-analog converter (DAC); 16-bit adjustable chip Internal analog-to-digital converter (ADC); high quality stereo earphone drive (30 ohms); separate analog, digital and IO power supplies; serial data and control interface (SPI);
8. External speaker, 3.5mm interface stereo headphone jack
9. Adjustable potentiometer, input analog signal
10. a 5-way joystick, a Reset button, a wakeup button, a Tamper button, a custom button
11. Voltmeter input interface
12. BNC input terminal, integrated dual channel CH1, CH2 oscilloscope circuit
13. 4 custom LEDs
14. RS232 serial communication interface (DB9)
15. MicroSD card slot (SD mode, no SD card), file system (FATFS)
16. Standard ARM JTAG 20 PIN emulation interface (easy to connect ST-LINK, JLINK, ULINK2 and other emulators)
17. USB HOST interface, embedded master/slave device controller (SL811HS), can communicate with USB devices at full speed or low speed.
18. LCD slot, support 3.5-inch 320*240 large screen 260,000-color TFT-LCD module, support 8/16-bit bus interface, mirror screen, ultra-high definition, FSMC control, also equipped with ADS7843 touch controller
19. External 2K IIC interface EEPROM (24LC02)
20. RS485 serial communication chip (SP3485)
21. CAN2.0A/B communication chip (SN65VHD230)
22. CAN2.0A/B communication interface, RS485 serial communication interface
23. 8M external quartz crystal
24. 32.768KHz quartz crystal oscillator, providing RTC clock
25. BOOT selection bit, using switch mode, making it easier for users to use
26. 74HC139 address decoder
27. USB2.0 full speed DEVICE interface

Each board comes with uCOSII + UCGUI demo program, and the CD comes with the source code of the compiled uCOSII + UCGUI routine, allowing you to easily enter the embedded world!

Product Image



Supporting materials

The following is included in the product package CD:

- install software
 - Contains a variety of good and easy to use software, is a tool for development
- Driver library
 - The official underlying drivers for each peripheral make development faster
- Experimental routine
 - Contains a large number of routines based on board hardware content
- Documentation
 - The datasheet of each device gives you more details
- Learning materials
 - Get an excellent introduction to STM32, so you can get started quickly
- Application document
 - ST's official application documents provide more in-depth development of STM32
- Schematic

Detailed configuration

1. RB-STM32F103 development board x1
2. 3.5 inch large screen color LCD module x1
3. USB slave adapter cable x1
4. Serial download line x1
5. RJ-45 network cable x1
6. Oscilloscope probe x1
7. 5V power adapter x1
8. Product CD (including programming examples, learning materials, software) x1



NEO-6

u-blox 6 GPS Modules

Data Sheet

Abstract

Technical data sheet describing the cost effective, high-performance u-blox 6 based NEO-6 series of GPS modules, that brings the high performance of the u-blox 6 positioning engine to the miniature NEO form factor.

These receivers combine a high level of integration capability with flexible connectivity options in a small package. This makes them perfectly suited for mass-market end products with strict size and cost requirements.



16.0 x 12.2 x 2.4 mm

www.u-blox.com

Document Information	
Title	NEO-6
Subtitle	u-blox 6 GPS Modules
Document type	Data Sheet
Document number	GPS.G6-HW-09005-E
Document status	

Document status information	
Objective Specification	This document contains target values. Revised and supplementary data will be published later.
Advance Information	This document contains data based on early testing. Revised and supplementary data will be published later.
Preliminary	This document contains data from product verification. Revised and supplementary data may be published later.
Released	This document contains the final product specification.

This document applies to the following products:

Name	Type number	ROM/FLASH version	PCN reference
NEO-6G	NEO-6G-0-001	ROM7.03	UBX-TN-11047-1
NEO-6Q	NEO-6Q-0-001	ROM7.03	UBX-TN-11047-1
NEO-6M	NEO-6M-0-001	ROM7.03	UBX-TN-11047-1
NEO-6P	NEO-6P-0-000	ROM6.02	N/A
NEO-6V	NEO-6V-0-000	ROM7.03	N/A
NEO-6T	NEO-6T-0-000	ROM7.03	N/A

This document and the use of any information contained therein, is subject to the acceptance of the u-blox terms and conditions. They can be downloaded from www.u-blox.com.

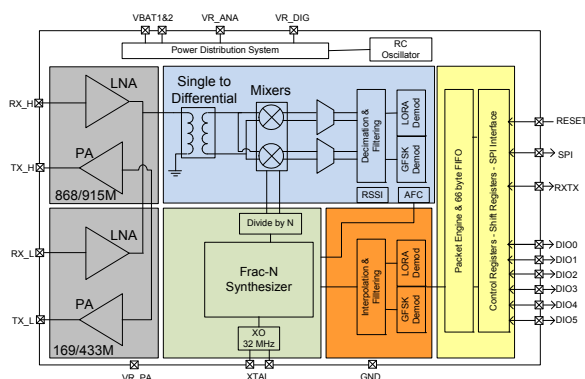
u-blox makes no warranties based on the accuracy or completeness of the contents of this document and reserves the right to make changes to specifications and product descriptions at any time without notice. Reproduction, use or disclosure to third parties without express permission is strictly prohibited. Copyright © 2011, u-blox AG.

u-blox® is a registered trademark of u-blox Holding AG in the EU and other countries. ARM® is the registered trademark of ARM Limited in the EU and other countries.



WIRELESS & SENSING

SX1276/77/78 - 137-1050 MHz Ultra Low Power Long Range Transceiver



GENERAL DESCRIPTION

Semtech's SX1276/77/78 family provides ultra long range while maintaining low current consumption, making it optimal for numerous applications.

With Semtech's patented modulation technique the device can achieve sensitivity of over -140 dBm using a low cost crystal and bill of materials. The high sensitivity combined with the integrated +20 dBm power amplifier creates the highest link budget making it optimal for any application requiring range. This modulation technique also provides significant advantages in blocking and selectivity over FSK.

The devices also support high performance (G)FSK for WMBus, FCC Part 90, 15.4g, and other legacy modes. Semtech's technology overcomes a typical trade-off of performance versus current consumption. The devices deliver exceptional phase noise, selectivity, receiver linearity, and IIP3 for significantly lower current consumption than competing devices.

The devices cover the major ISM bands from 137 MHz to 1050 MHz. For ease of matching, low BOM cost, and higher performance the SX1276/77/78 family offers separate high band and low band support for applications wanting to cover dual bands for world-wide operation, dual band support, or security against future band allocation changes.

KEY PRODUCT FEATURES

- ◆ 168 dB maximum link budget
- ◆ +20 dBm - 100 mW constant RF output vs. V supply
- ◆ +14 dBm high efficiency PA
- ◆ Programmable bit rate up to 300 kbps
- ◆ High sensitivity: down to -146.5 dBm
- ◆ Bullet-proof front end: IIP3 = -12 dBm
- ◆ 100 dB blocking immunity
- ◆ Low RX current of 10 mA, 200 nA register retention
- ◆ Fully integrated synthesizer with a resolution of 61 Hz
- ◆ FSK, GFSK, MSK, GMSK, LORA and OOK modulations
- ◆ Built-in bit synchronizer for clock recovery
- ◆ Sync word recognition
- ◆ Preamble detection
- ◆ 115 dB+ Dynamic Range RSSI
- ◆ Automatic RF Sense with ultra-fast AFC
- ◆ Packet engine up to 64 bytes with CRC
- ◆ Built-in temperature sensor and Low Battery indicator

APPLICATIONS

- ◆ Automated Meter Reading
- ◆ Wireless Sensor Networks
- ◆ Home and Building Automation
- ◆ Wireless Alarm and Security Systems
- ◆ Industrial Monitoring and Control

ORDERING INFORMATION

Part Number	Delivery	MOQ / Multiple
SX1276	T&R	3000 pieces
SX1277	T&R	3000 pieces
SX1278	T&R	3000 pieces

DOIT Esp32 DevKit v1

The DOIT Esp32 DevKit v1 is one of the development board created by DOIT to evaluate the ESP-WROOM-32 module. It is based on the [ESP32 microcontroller](#) that boasts Wifi, Bluetooth, Ethernet and Low Power support all in a single chip.



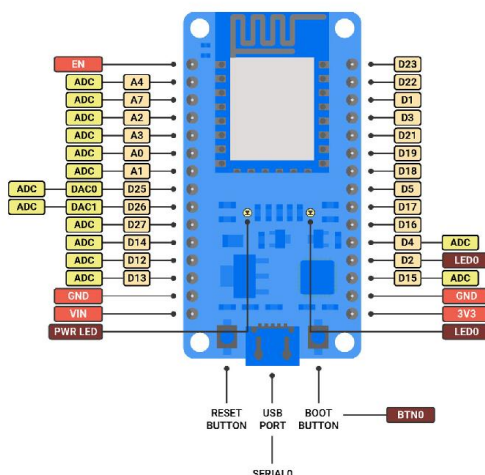
Pin Mapping

DO NOT USE D6 TO D11

PWM IS ENABLED ON EVERY DIGITAL PIN

ICU NOT SUPPORTED

ADC ON PINS D4, D12, D13, D14, D15, D25, D26, D27
CAN BE READ ONLY WITH WI-FI NOT STARTED



More info about DOIT Esp32 DevKit v1 can be found [here](#).

Flash Layout

The internal flash of the ESP32 module is organized in a single flash area with pages of 4096 bytes each. The flash starts at address 0x000000, but many areas are reserved for Esp32 IDF SDK and Zerynth VM. There exist two different layouts based on the presence of BLE support.

In particular, for non-BLE VMs:

Start address	Size	Content
0x00009000	16Kb	Esp32 NVS area
0x0000D000	8Kb	Esp32 OTA data
0x0000F000	4Kb	Esp32 PHY data
0x00010000	1Mb	Zerynth VM
0x00110000	1Mb	Zerynth VM (FOTA)
0x00210000	512Kb	Zerynth Bytecode
0x00290000	512Kb	Zerynth Bytecode (FOTA)
0x00310000	512Kb	Free for user storage
0x00390000	448Kb	Reserved

For BLE VMs:

Start address	Size	Content
0x00009000	16Kb	Esp32 NVS area
0x0000D000	8Kb	Esp32 OTA data
0x0000F000	4Kb	Esp32 PHY data
0x00010000	1216Kb	Zerynth VM
0x00140000	1216Kb	Zerynth VM (FOTA)
0x00270000	320Kb	Zerynth Bytecode
0x002C0000	320Kb	Zerynth Bytecode (FOTA)
0x00310000	512Kb	Free for user storage
0x00390000	448Kb	Reserved

Device Summary

- Microcontroller: Tensilica 32-bit Single-/Dual-core CPU Xtensa LX6
- Operating Voltage: 3.3V
- Input Voltage: 7-12V
- Digital I/O Pins (DIO): 25
- Analog Input Pins (ADC): 6
- Analog Outputs Pins (DAC): 2
- UARTs: 3
- SPIs: 2
- I2Cs: 3
- Flash Memory: 4 MB
- SRAM: 520 KB
- Clock Speed: 240 Mhz
- Wi-Fi: IEEE 802.11 b/g/n/e/i:
 - Integrated TR switch, balun, LNA, power amplifier and matching network
 - WEP or WPA/WPA2 authentication, or open networks

Power

Power to the DOIT Esp32 DevKit v1 is supplied via the on-board USB Micro B connector or directly via the “VIN” pin. The power source is selected automatically.

The device can operate on an external supply of 6 to 20 volts. If using more than 12V, the voltage regulator may overheat and damage the device. The recommended range is 7 to 12 volts.

Connect, Register, Virtualize and Program

The DOIT Esp32 DevKit v1 comes with a serial-to-usb chip on board that allows programming and opening the UART of the ESP32 module. Drivers may be needed depending on your system (Mac or Windows) and can be download from the official [Espressif documentation](#) page. In Linux systems, the DevKit v1 should work out of the box.

Note

For Linux Platform: to allow the access to serial ports the user needs read/write access to the serial device file. Adding the user to the group, that owns this file, gives the required read/write access:

- **Ubuntu** distribution → dialout group
- **Arch Linux** distribution → uucp group

Once connected on a USB port, if drivers have been correctly installed, the DevKit v1 device is recognized by Zerynth Studio. The next steps are:

- **Select** the DevKit v1 on the **Device Management Toolbar** (disambiguate if necessary);
- **Register** the device by clicking the “Z” button from the Zerynth Studio;
- **Create** a Virtual Machine for the device by clicking the “Z” button for the second time;
- **Virtualize** the device by clicking the “Z” button for the third time.

Note

No user intervention on the device is required for registration and virtualization process

After virtualization, the DevKit v1 is ready to be programmed and the Zerynth scripts **uploaded**. Just **Select** the virtualized device from the “Device Management Toolbar” and **click** the dedicated “upload” button of Zerynth Studio.

Note

No user intervention on the device is required for the uplink process.

Firmware Over the Air update (FOTA)

The Firmware Over the Air feature allows to update the device firmware at runtime. Zerynth FOTA in the DevKitC device is available for bytecode and VM.

Flash Layout is shown in table below:

Start address	Size	Content
0x00010000	1Mb	Zerynth VM (slot 0)
0x00110000	1Mb	Zerynth VM (slot 1)
0x00210000	512Kb	Zerynth Bytecode (slot 0)
0x00290000	512Kb	Zerynth Bytecode (slot 1)

For BLE VMs:

Start address	Size	Content
0x00010000	1216Kb	Zerynth VM (slot 0)
0x00140000	1216Kb	Zerynth VM (slot 1)

Start address	Size	Content
0x00270000	320Kb	Zerynth Bytecode (slot 0)
0x002C0000	320Kb	Zerynth Bytecode (slot 1)

For Esp32 based devices, the FOTA process is implemented mostly by using the provided system calls in the IDF framework. The selection of the next VM to be run is therefore a duty of the Espressif bootloader; the bootloader however, does not provide a failsafe mechanism to revert to the previous VM in case the currently selected one fails to start. At the moment this lack of a safety feature can not be circumvented, unless by changing the bootloader. As soon as Espressif relases a new IDF with such feature, we will release updated VMs.

Secure Firmware

Secure Firmware feature allows to detect and recover from malfunctions and, when supported, to protect the running firmware (e.g. disabling the external access to flash or assigning protected RAM memory to critical parts of the system).

This feature is strongly platform dependent; more information at [Secure Firmware - ESP32 section](#).

Zerynth Secure Socket

To be able to use Zerynth Secure Socket on esp32 boards `NATIVE_MBEDTLS: true` must be used instead of `ZERYNTH_SSL: true` in the `project.yml` file.

Missing features

Not all IDF features have been included in the Esp32 based VMs. In particular the following are missing but will be added in the near future:

- Touch detection support

Appendix B

Codes

TRANSMITTER

```
#include <SPI.h>

#include <LoRa.h>

#include <TinyGPS++.h>

#include <SoftwareSerial.h>

#define LORA_SS 10

#define LORA_RST 9

#define LORA_DIO0 2

#define GPS_RX 4

#define GPS_TX 3

#define USER_BUTTON PC13

SoftwareSerial gpsSerial(GPS_RX, GPS_TX); // RX, TX

TinyGPSPlus gps;

void setup() {

  Serial.begin(115200);

  gpsSerial.begin(9600);

  pinMode(USER_BUTTON, INPUT);


  Serial.println("GPS + LoRa Transmitter Initializing...");


  LoRa.setPins(LORA_SS, LORA_RST, LORA_DIO0);

  if (!LoRa.begin(433E6)) { // 433 MHz

    Serial.println("Starting LoRa failed!");

    while (1);

  }

  Serial.println("LoRa Initialized.");
}

void loop() {

  while (gpsSerial.available() > 0) {
```

```

    gps.encode(gpsSerial.read());
}

if (gps.location.isUpdated()) {
    float lat = gps.location.lat();
    float lon = gps.location.lng();

    // Print to Serial Monitor
    Serial.print("Lat: "); Serial.print(lat, 6);
    Serial.print(" | Lon: "); Serial.println(lon, 6);

    String message;

    if (digitalRead(USER_BUTTON) == LOW) {
        Serial.println("🚨 EMERGENCY BUTTON PRESSED!");
        message = "EMERGENCY," + String(lat, 6) + "," + String(lon, 6);
    } else {
        // Default GPS transmission
        message = String(lat, 6) + "," + String(lon, 6);
    }

    LoRa.beginPacket();
    LoRa.print(message);
    LoRa.endPacket();

    Serial.println("Sent via LoRa: " + message);
    delay(1000);
} }

```


RECEIVER

```
#include <SPI.h>

#include <LoRa.h>

#define LORA_SCK 18

#define LORA_MISO 19

#define LORA_MOSI 23

#define LORA_SS 5

#define LORA_RST 14

#define LORA_DIO0 2

void setup() {

    Serial.begin(115200);

    Serial.println("LoRa Receiver Starting...");

    SPI.begin(LORA_SCK, LORA_MISO, LORA_MOSI, LORA_SS);

    Serial.println("LoRa Receiver Initialized on 433 MHz");

}

void loop() {

    int packetSize = LoRa.parsePacket();

    if (packetSize) {

        String incoming = "";

        while (LoRa.available()) {

            incoming += (char)LoRa.read();

        }

        Serial.print("Received: ");

        Serial.println(incoming);

        int firstComma = incoming.indexOf(',');

        int secondComma = incoming.indexOf(',', firstComma + 1);

        if (firstComma > 0 && secondComma == -1) {

            float lat = incoming.substring(0, firstComma).toFloat();

            float lon = incoming.substring(firstComma + 1).toFloat();
```

```
Serial.print("Latitude: "); Serial.println(lat, 6);

Serial.print("Longitude: "); Serial.println(lon, 6);
}

else if (firstComma > 0 && secondComma > 0) {

    String alert = incoming.substring(0, firstComma);

    float lat = incoming.substring(firstComma + 1, secondComma).toFloat();

    float lon = incoming.substring(secondComma + 1).toFloat();

    Serial.println(" 🚨 EMERGENCY ALERT RECEIVED 🚨 ");

    Serial.print("Latitude: "); Serial.println(lat, 6);

    Serial.print("Longitude: "); Serial.println(lon, 6);

}

Serial.println("-----");

}

}
```